

Introduction to Community Detection in Networks via Modularity Maximization

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Introduction

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 - ◇ the connections may also have attributes such as hyperlink, road, and shipping lane.

Example: Web Pages

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Releases

Stable (notes)
2.8.8 – November 2022
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Latest (notes)
3.0 development
[github](#) | [doc](#) | [pdf](#)

[Archive](#)



NetworkX is a Python package for the creation, manipulation, and study of the structure, dynamics, and functions of complex networks.

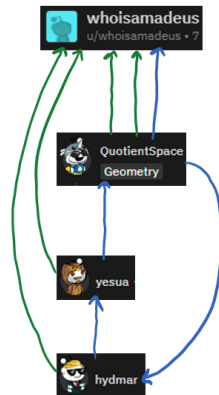
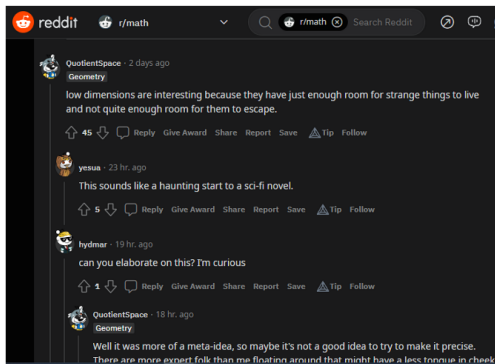
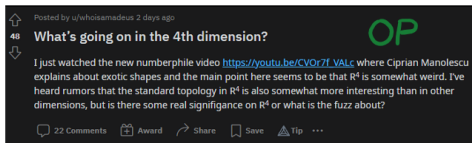


Software for complex networks

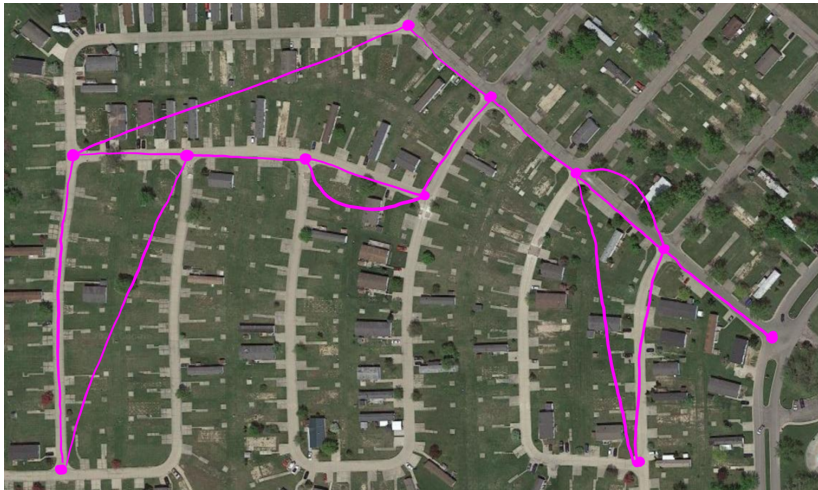
- Data structures for graphs, digraphs, and multigraphs
- Many standard graph algorithms
- Network structure and analysis measures



Example: Social Media



Example: Transit



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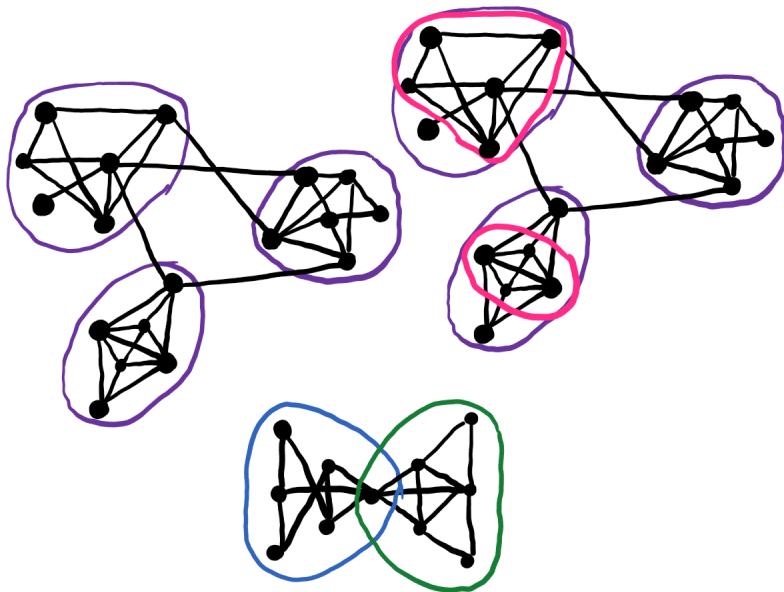
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 - ◇ often communities are considered to have a high density of internal edges compared to that of external edges.

Community Examples



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 - ◇ etc.

Methods

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- ▶ Maximum Likelihood Estimation.

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Today we will focus on Modularity Maximization methods.

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In general different models generate graphs based on an underlying theory of how edges are formed between nodes. Modularity Maximization leverages the configuration model.

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- ▶ The number of ways to form an edge from node i to node j in a directed network is $k_i^{out} k_j^{in}$.

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- ▶ $\vec{\mathbb{E}}_{i,j} = \frac{k_i^{out} k_j^{in}}{m}.$

Proof of Spoke Expectation (Undirected)

$$\sum_{i,j}^n \mathbb{E}_{i,j} = \sum_{i,j \neq i}^n \mathbb{E}_{i,j} + \sum_i^n \mathbb{E}_{i,i}$$

Proof of Spoke Expectation (Undirected)

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$$Q = \frac{1}{s} \left[\sum_{i,j \neq i}^n \left(A_{i,j} - \frac{k_i k_j}{s-1} \right) \delta(c_i, c_j) + \sum_i^n \left(A_{i,i} - \frac{k_i(k_i-1)}{s-1} \right) \right].$$

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In an iterative fashion, we can performed the reassignment with the greatest increase in modularity.

One note on modularity maximization methods is that they detect disjoint communities, so it has a decreased accuracy for networks with underlying hierarchical or overlapping mechanics.

Complexity Optimization in Modularity Maximization [1/2]

In general, modularity maximization techniques are often used due to their ability to analyze massive networks (millions of nodes).

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- ▶ How we reassign communities.
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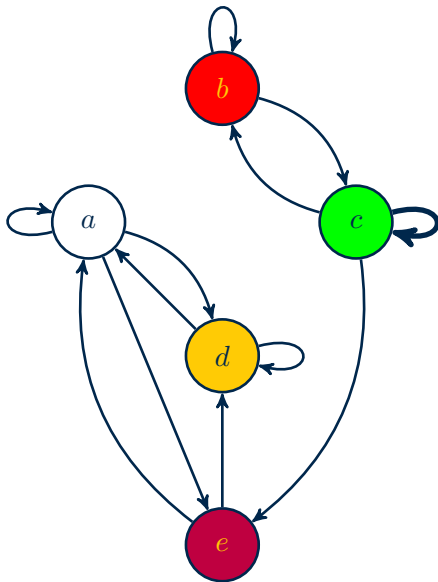
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- ▶ Cache relevant values (node degrees, expectations) rather than computing during runtime.
- ▶ Halt the algorithm once the highest possible change in modularity is below a set threshold.

Example: Network



Example: Adjacency Matrix

$\vec{A}$	a	b	c	d	e
a	1	0	0	1	1
b	0	1	1	0	0
c	0	1	2	0	1
d	1	0	0	1	0
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The adjacency matrix is the expected form computers will store networks in.

Example: Out-Degree Calculations

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$$k_a^{in} = 1 + 0 + 0 + 1 + 1 = 3.$$

Example: All Degree Calculations

i	a	b	c	d	e
k_i^{out}	3	2	4	2	2
k_i^{in}	3	2	3	3	2
k_i	6	4	7	5	4

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$$\vec{Q} = \frac{1}{m} \left[(\vec{A}_{a,a} - \vec{\mathbb{E}}_{a,a}) + (\vec{A}_{b,b} - \vec{\mathbb{E}}_{b,b}) \right. \\ \left. + (\vec{A}_{c,c} - \vec{\mathbb{E}}_{c,c}) + (\vec{A}_{d,d} - \vec{\mathbb{E}}_{d,d}) + (\vec{A}_{e,e} - \vec{\mathbb{E}}_{e,e}) \right].$$

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$$\vec{Q} = \frac{1}{13} \left[\left(1 - \frac{3 \cdot 3}{13}\right) + \left(1 - \frac{2 \cdot 2}{13}\right) \right. \\ \left. + \left(2 - \frac{4 \cdot 3}{13}\right) + \left(1 - \frac{2 \cdot 3}{13}\right) + \left(0 - \frac{2 \cdot 2}{13}\right) \right].$$

Example: Modularity Calculation [2/2]

$$\vec{Q} = \frac{1}{13} \left[\frac{4}{13} + \frac{9}{13} + \frac{14}{13} + \frac{7}{13} - \frac{4}{13} \right].$$

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By doing so we can treat merged nodes as meta nodes, and form a new adjacency matrix with them.

As a result we need only update the cached values which change.

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For our adjacency matrix this is just a row and column merge.

Similarly, for recalculating k_i^{out} and k_i^{in} we need only one addition.

Last, we will only consider merges for nodes which are neighbors.

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The merges we need consider should be much less than the $\binom{n}{2}$ node pairs.

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We will choose the node merge with the best change $\Delta\vec{Q}$.

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<i>merge</i>	$A_{i,i}$	$A_{j,j}$	$A_{i,j}$	$A_{j,i}$	$A_{i',i'}$
$\{a, d\}$	1	1	1	1	4
$\{a, e\}$	1	0	1	1	3
$\{b, c\}$	1	2	1	1	5
$\{c, e\}$	2	0	1	0	3
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<i>merge</i>	k_i^{out}	k_j^{out}	$k_{i'}^{out}$	k_i^{in}	k_j^{in}	$k_{i'}^{in}$
$\{a, d\}$	3	2	5	3	3	6
$\{a, e\}$	3	2	5	3	2	5
$\{b, c\}$	2	4	6	2	3	5
$\{c, e\}$	4	2	6	3	2	5
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Where i' is the new meta node formed from merging nodes i and j .

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$\{a, d\}$	4/13	7/13	22/13	11/169
$\{a, e\}$	4/13	-4/13	14/13	14/169
$\{b, c\}$	9/13	14/13	35/13	12/169
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The process should continue until the best change in modularity is lower than a set threshold.

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Let us perform the best merge for each node.

<i>merge</i>	$\partial \vec{Q}_i$	$\partial \vec{Q}_j$	$\partial \vec{Q}_{i'}$	$\Delta \vec{Q}$
$\{a, d\}$	4/13	7/13	22/13	11/169
$\{a, e\}$	4/13	-4/13	14/13	14/169
$\{b, c\}$	9/13	14/13	35/13	12/169
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Example: Multiple Merges [1/2]

To minimize complexity, one may perform many merges in a single iteration.

As is, we already compute modularity gain for each considered merge.

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The best merge for nodes a , and e , is to merge them together, similarly, the best for b , and c , is to merge them together, last, the best merge for d is to merge it with a .

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For a small network as in the example, we detect the underlying community structure in one iteration.

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Thus nodes a, d, e form one community, and nodes b, c another.

Naive Complexity [1/4]

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Computing modularity alone is quite intensive, and how about computing the change in modularity?

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 - ◇ Do such for each possible merge: $\mathcal{O}(mn^2)$.

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Naive Complexity [4/4]

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- ▶ Total community detection: $\mathcal{O}(mn^3)$.

Naive Complexity [4/4]

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- ▶ Total community detection: $\mathcal{O}(mn^3)$.
- ◊ Calculate best $\Delta \vec{Q}_i$: $\mathcal{O}(mn^2)$.

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- ▶ Total community detection: $\mathcal{O}(mn^3)$.
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- * Calculate all $\Delta \vec{Q}_i$.

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Computing modularity alone is quite intensive, and how about computing the change in modularity?

- ▶ Total community detection: $\mathcal{O}(mn^3)$.
 - ◇ Calculate best $\Delta \vec{Q}_i : \mathcal{O}(mn^2)$.
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 - * $\mathcal{O}(n + mn^2) = \mathcal{O}(mn^2)$.

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Conclusion

We truly covered a lot, and yet this is only a glimpse of community detection algorithms.

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Papers are still coming out on the topic, and new methods are being invented.

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QUESTIONS?

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